

NABINDRA RAJ SHRESTHA



Nuwakot, Nepal



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GitHub



Google Scholar



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EDUCATION

Nov 2017 – Apr 2022

Bachelor of Electrical Engineering

Tribhuvan University, Institute of Engineering (IOE), Pulchowk Campus

Relevant Coursework: Power System Analysis, Power Electronics, Electrical Machines, Control System, Digital Control System, Engineering Mathematics, Object Oriented Programming, Electric Circuit Theory, Electronic Devices and Circuits, Digital Logic.



PUBLICATIONS

Conferences

[C1] **N. Shrestha**, P. Sherpa, S. Gurung, K. Chapagain, B. Mallik, B. Pokhrel, “**A Custom Developed Flexible Tool for Real Time Distribution Grid Visualization**”, 2024 IEEE 33rd Industrial Symposium on Industrial Electronics (ISIE), Ulsan, South Korea.

[C2] P. Sherpa, **N. Shrestha**, K. Chapagain, B. Pokhrel, S. Dahal, P. Bhattarai, S. Gurung, “**Real Time Distribution Grid Visualization using Smart Meters**”, 2024 IEEE 21st International Conference on Electrical Engineering, Computer Telecommunications, and Information Technology, Khon Kaen, Thailand.

Journals

[J1] P. Sherpa, **N. Shrestha**, S. Gurung, K. Chapagain, B. Pokhrel, B. Mallik, "Development of a Test-Bed for the Dynamic Visualization of the Distribution Grid". [Submitted for publication to IET Generation Transmission & Distribution]



PROFESSIONAL EXPERIENCE

Jan 2024 – Present

Research Assistant

DoEEE, Kathmandu university, Dhulikhel, Nepal [D-VA](#)

- Develop an open-source platform for real-time distribution grid visualization.
- Develop a predictive grid visualization tool implementing deep learning based load forecasting models.
- Study the suitability of various communication protocols used in distribution automation.

Nov 2022 – Dec 2023

Project Officer

People, Energy & Environment Development Association, Kathmandu, Nepal, [PEEDA](#)

- Collected and analyzed data on Micro-Hydropower (MHP) systems and transformers to assess their capacity for supporting electric cooking projects.
- Authored books and manuals on the repair of induction stoves and electric pressure cookers.
- Assisted in municipal energy planning (MEP) and facilitated the local government in various stages of MEP.
- Organized and led training programs on the repair of electric cooking appliances.

Jan 2022 – Aug 2022

Battery Management System Engineer

Royal Automobiles, Kathmandu, Nepal

- Developed equivalent circuit models of different Li-ion Cells.
- Designed the specifications and configuration of a 60-kWh battery pack.
- Developed a MATLAB model to simulate the battery pack under CCCV and CPCV conditions.
- Developed MATLAB models for the state of charge estimation of Li-ion cells.



SKILLS

Software & Programming

MATLAB & SIMULINK, DigSILENT PowerFactory, LaTeX, Python, C/C++, AutoCAD, Proteus

Packages/Frameworks

PyPSA, Pytorch, Scikit, Learn, Django, Folium, Numpy, Pandas, Matplotlib

ML/DL

Logistic Regression, Decision Tree, Random Forest, SVM, Neural Network

Hardware

Arduino, Raspberry Pi, Atmega 16/32



PROJECTS

AI Approach to Estimate the State of Charge of an Electric Vehicle Battery

Final year undergraduate thesis project. Developed different machine-learning models to estimate the state of charge of an electric vehicle battery and analyzed the performance of each model. Implemented the best-performing model, Random Forest, on hardware.

Prepaid Energy Meter

Third-year group project. Converted an electromechanical induction-type energy meter to facilitate prepaid services based on GSM technology.

Circuit Simulator

Second year computer programming project. Built a GUI-based basic electrical circuit simulator in C++.

Let Us Calculate

First-year computer programming project. Built a program in C to make a calculator that could perform a variety of operations – from solving simple algebraic expressions to matrix operations, quadratic equations, and the system of linear equations.



AWARDS AND CERTIFICATIONS

LOCUS 2020 WINNER/ Electrical Category

Prepaid Energy Meter won the title of *The Best Electrical Project* in the LOCUS 2020 – the biggest technological festival in Nepal.

Certification of Energy Audit Training

Completed a comprehensive 5-day energy audit training program, building capacity in energy efficiency assessments, and identifying energy-saving opportunities.